

KON 313

Feedback Control Systems

Modeling of the Electro-Mechanical Systems in Frequency Domain

Dr. Aysa Jafari Farmand

aysa.jafari@itu.edu.tr

Contents

- Electromechanical Systems
- DC Servomotor

Electromechanical Systems

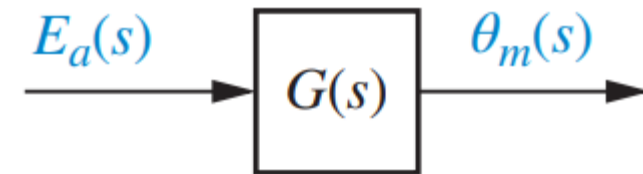
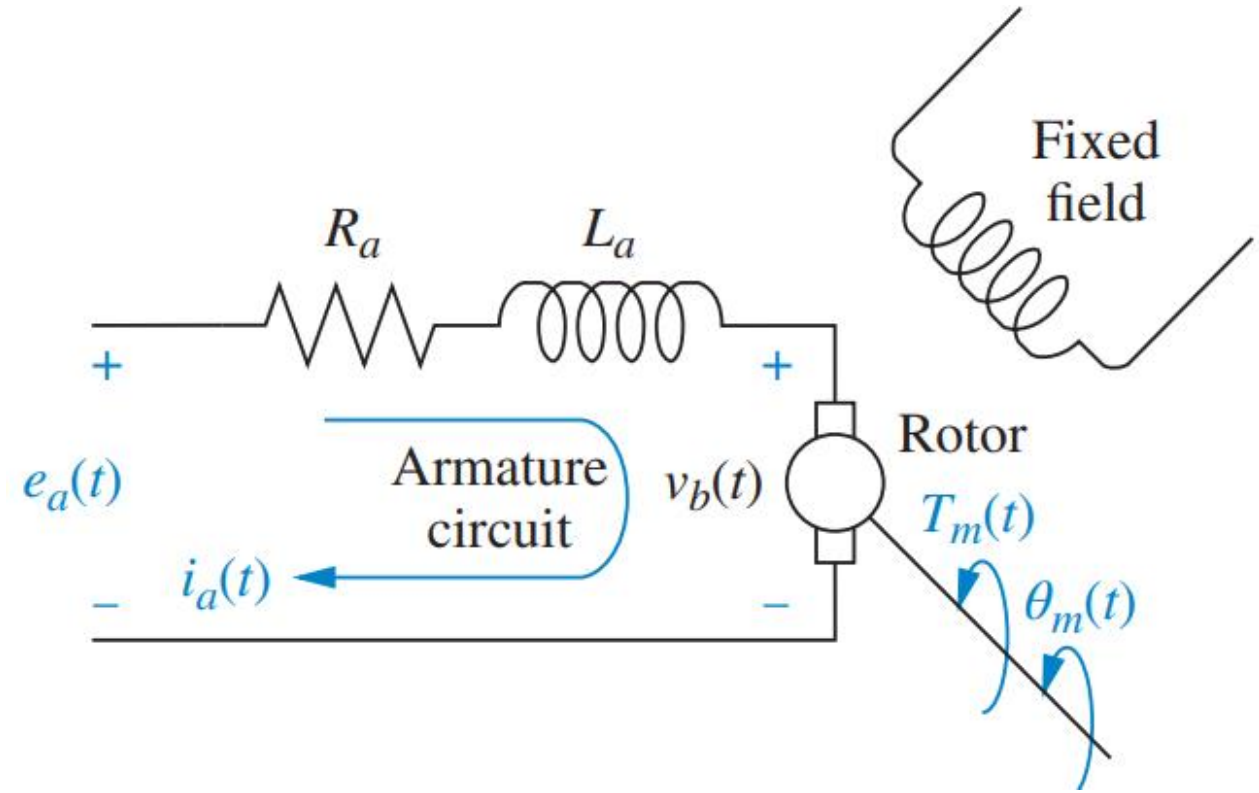
- *Electromechanical* systems are hybrids of electrical and mechanical variables, like robot controls.
- A motor is an electromechanical component that yields a displacement output for a voltage input, that is, a mechanical output generated by an electrical input.
- Armature-controlled **dc servomotor** is a particular kind of electromechanical system.

DC Servomotor

A magnetic field is developed by stationary permanent magnets or a stationary electromagnet called the *fixed field*.

A rotating circuit called the *armature*, through which current $i_a(t)$ flows, passes through this magnetic field at right angles.

- It feels a force and the rotor turns.
- generates a voltage at the terminals of the conductor.



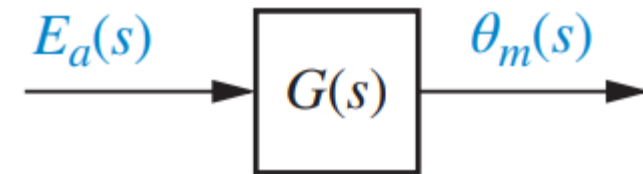
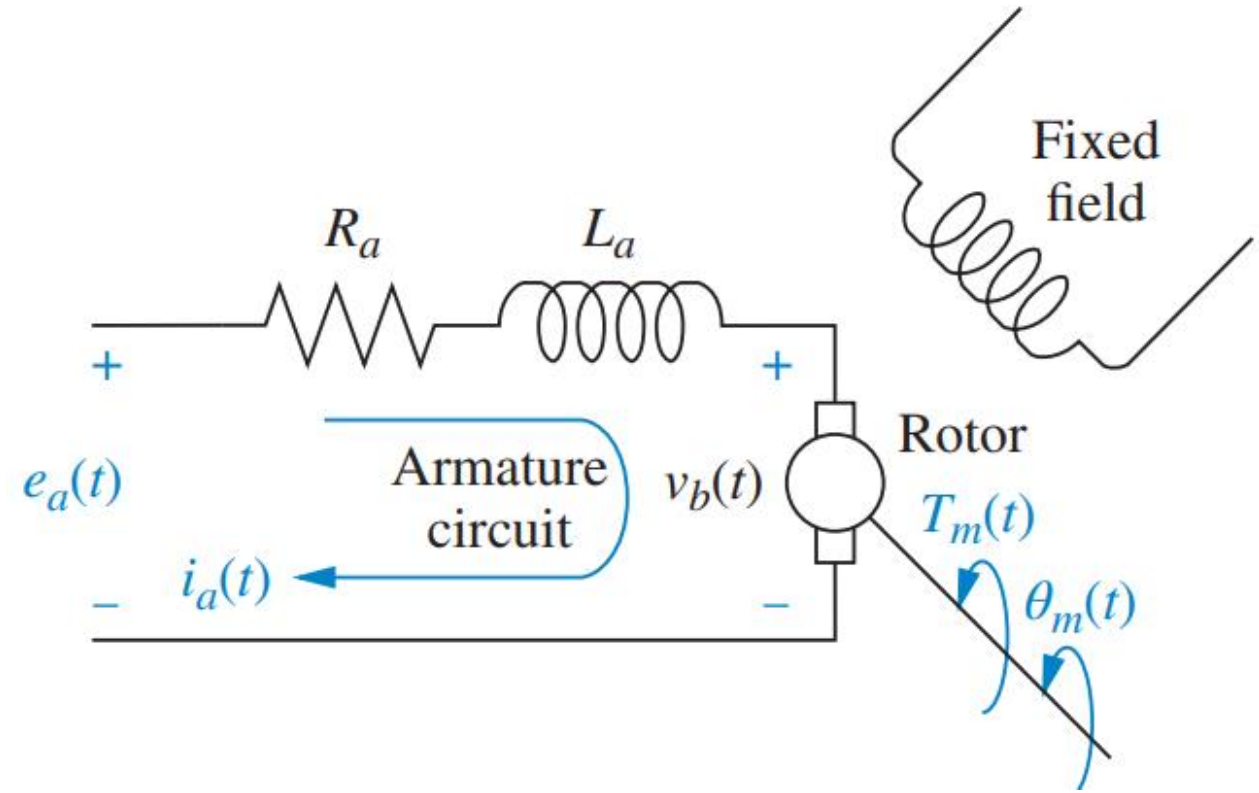
DC Servomotor

Armature feels a force, $F = Bli_a(t)$ where B is the magnetic field strength and l is the length of the conductor.

The resulting torque, T_m , turns the *rotor*, the rotating part of the motor.

$$T_m(s) = K_t I_a(s)$$

where K_t is called the motor torque constant, which depends on the motor and magnetic field characteristics.



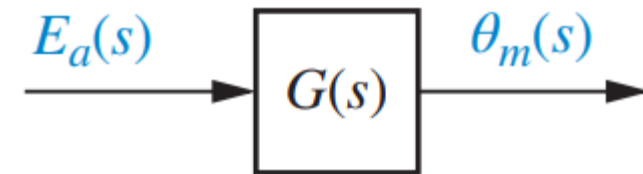
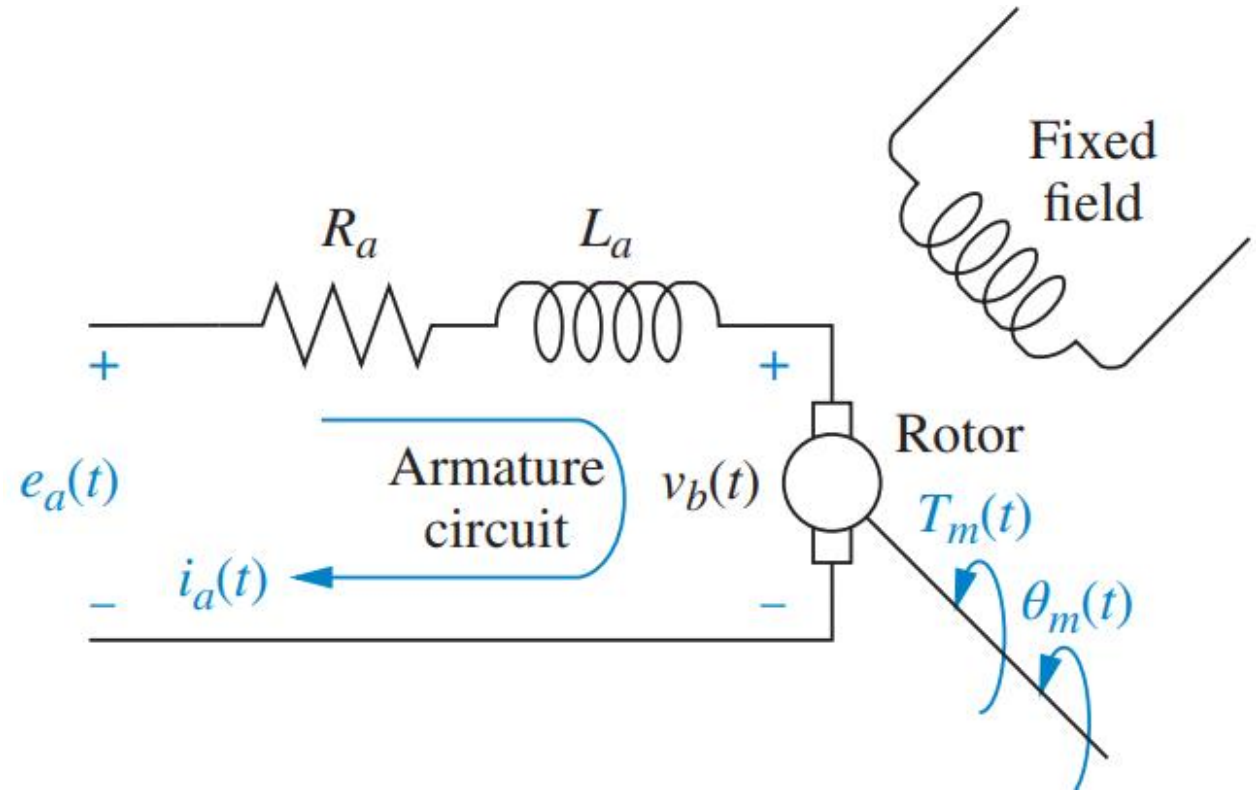
DC Servomotor

Since the current-carrying armature is rotating in a magnetic field, its voltage is proportional to speed.

$$v_b(t) = K_b \frac{d\theta_m(t)}{dt}$$

We call $v_b(t)$ the back electromotive force (back emf); K_b is called the back emf constant; and $\frac{d\theta_m(t)}{dt}$ is the angular velocity of the motor. Taking the Laplace transform,

$$V_b(s) = K_b s \theta_m(s)$$



DC Servomotor

Writing KVL around the Laplace transformed armature circuit:

$$R_a I_a(s) + L_a s I_a(s) + V_b(s) = E_a(s)$$

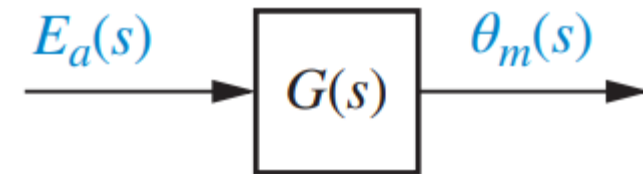
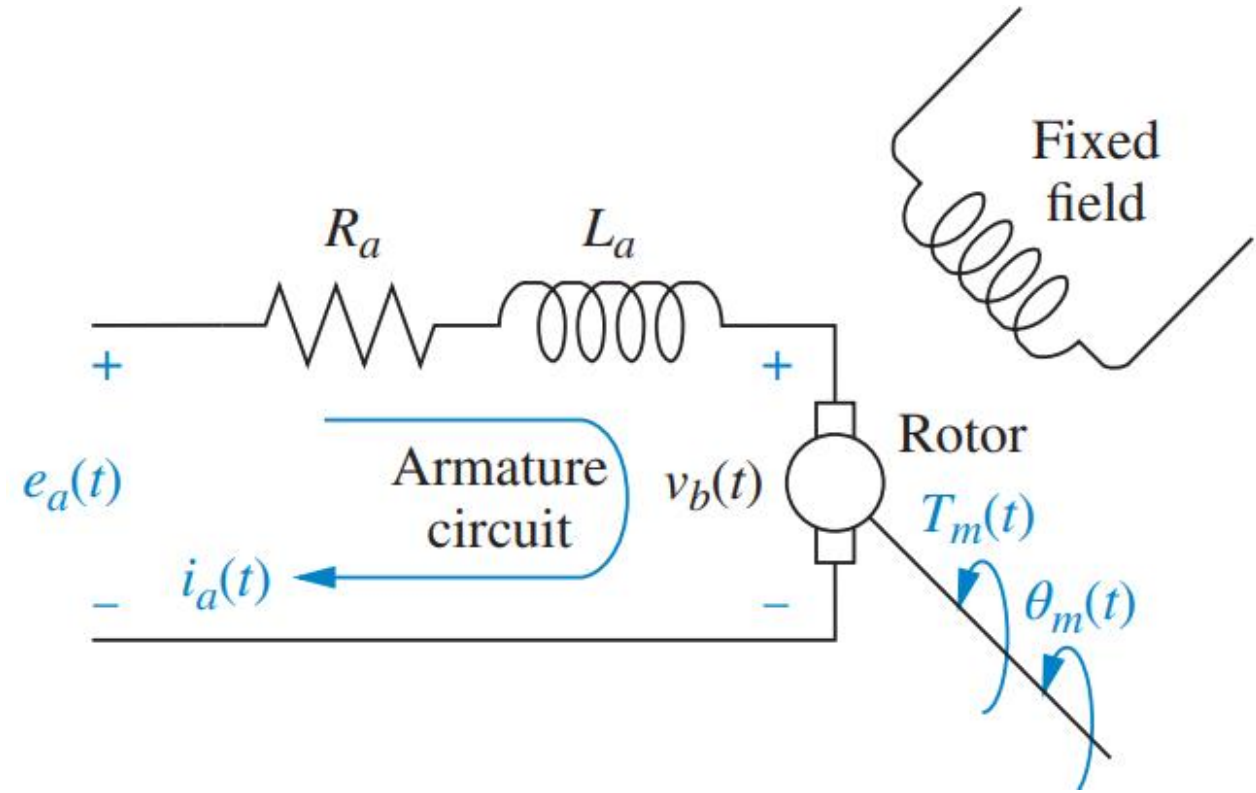
We had:

$$V_b(s) = K_b s \theta_m(s)$$

$$I_a(s) = \frac{1}{K_t} T_m(s)$$

Thus:

$$\frac{(R_a + L_a s) T_m(s)}{K_t} + K_b s \theta_m(s) = E_a(s)$$



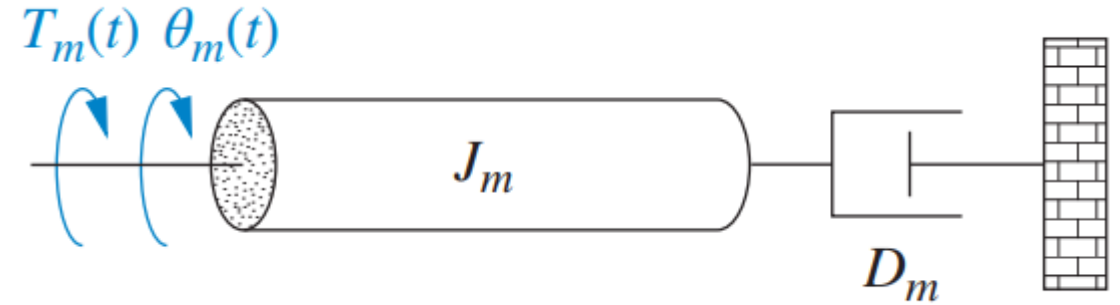
DC Servomotor

Considering a typical equivalent mechanical loading on a motor

$$T_m(s) = (J_ms^2 + D_ms)\theta_m(s)$$

J_m : Equivalent inertia (armature and load)

D_m : Equivalent viscous damping (armature and load)



$$\frac{(R_a + L_as)(J_ms^2 + D_ms)\theta_m(s)}{K_t} + K_b s\theta_m(s) = E_a(s)$$

The block diagram shows the transfer function $G(s)$ of the motor. The input is $E_a(s)$ and the output is $\theta_m(s)$.

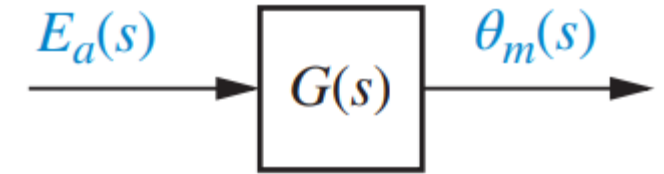
For a dc motor, the armature inductance, L_a , is usually small compared to the armature resistance, R_a , thus

$$\left[\frac{R_a}{K_t} (J_ms + D_m) + K_b \right] s\theta_m(s) = E_a(s)$$

DC Servomotor

The transfer function:

$$\frac{\theta_m(s)}{E_a(s)} = \frac{K_t / (R_a J_m)}{s \left[s + \frac{1}{J_m} \left(D_m + \frac{K_t K_b}{R_a} \right) \right]}$$



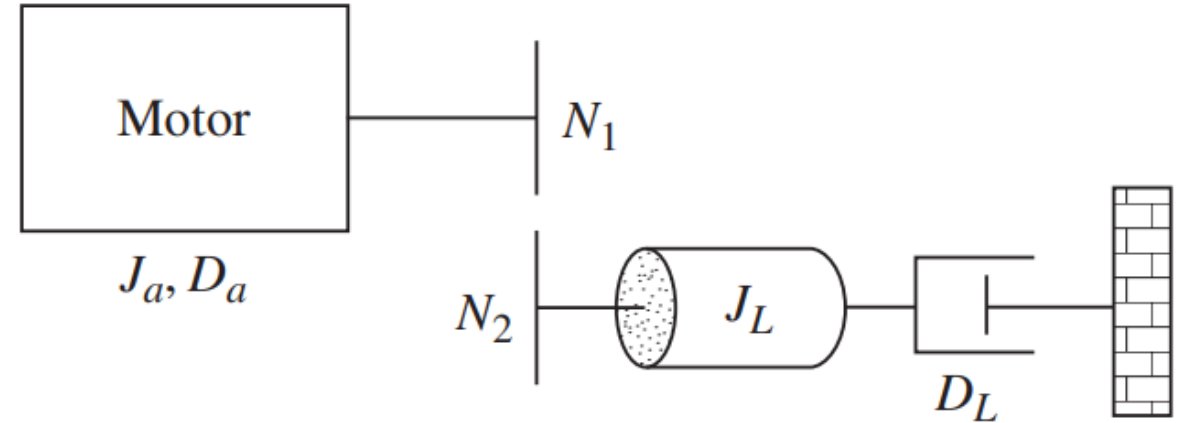
Namely:

$$\frac{\theta_m(s)}{E_a(s)} = \frac{K}{s(s + \alpha)}$$

DC Servomotor

The mechanical constants, J_m and D_m :

- Consider a motor with inertia J_a and damping D_a at the armature driving a load consisting of inertia J_L and damping D_L .



- Reflecting all inertia and damping values back to the armature, the equivalent values at the armature are

$$J_m = J_a + J_L \left(\frac{N_1}{N_2} \right)^2 ; \quad D_m = D_a + D_L \left(\frac{N_1}{N_2} \right)^2$$

DC Servomotor

The electrical constants, $\frac{R_a}{K_t}$ and K_b :

- These constants can be obtained through a *dynamometer* test of the motor; A dynamometer measures the torque and speed of a motor under the condition of a **constant applied voltage**.
- Considering small value for inductance, L_a we had:

$$\frac{R_a}{K_t} T_m(s) + K_b s \theta_m(s) = E_a(s) \quad \xRightarrow{\mathcal{L}^{-1}} \quad \frac{R_a}{K_t} T_m(t) + K_b \omega_m(t) = e_a(t)$$

Hence, if a dc voltage, e_a , is applied, the motor will turn at a constant angular velocity, ω_m , with a constant torque, T_m . operating at steady state:

$$\frac{R_a}{K_t} T_m + K_b \omega_m = e_a$$

DC Servomotor

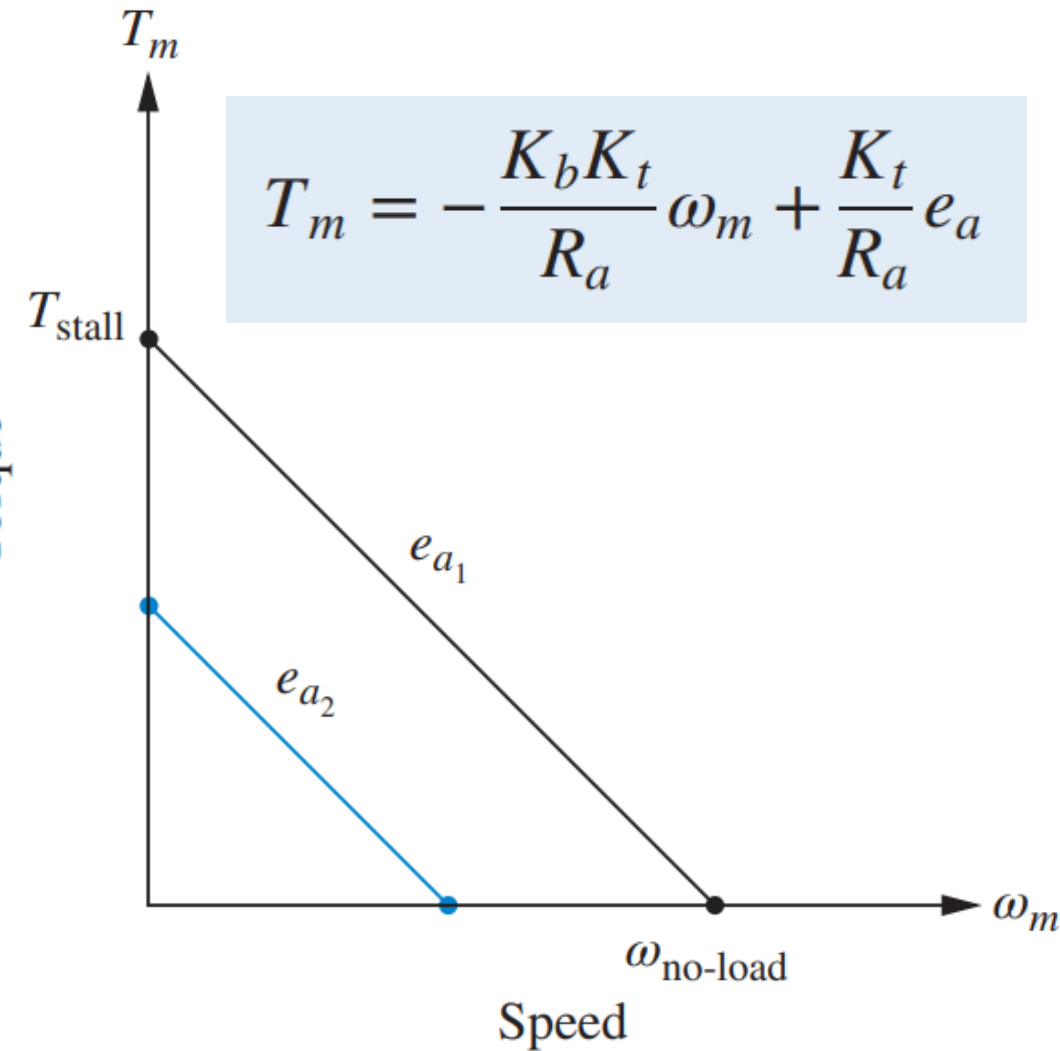
The electrical constants, $\frac{R_a}{K_t}$ and K_b :

The *torque-speed curve*.

- *Stall torque*, T_{stall} : The torque axis interception point when the angular velocity reaches zero.
- *No-load speed*, $\omega_{no-load}$: The angular velocity occurring when the torque is zero.

The electrical constants can be found from a dynamometer test of the motor, which yield T_{stall} and $\omega_{no-load}$ for a given e_a .

Torque

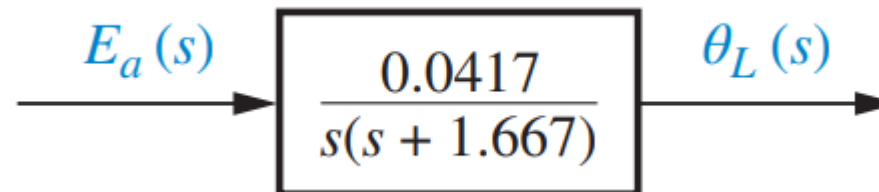
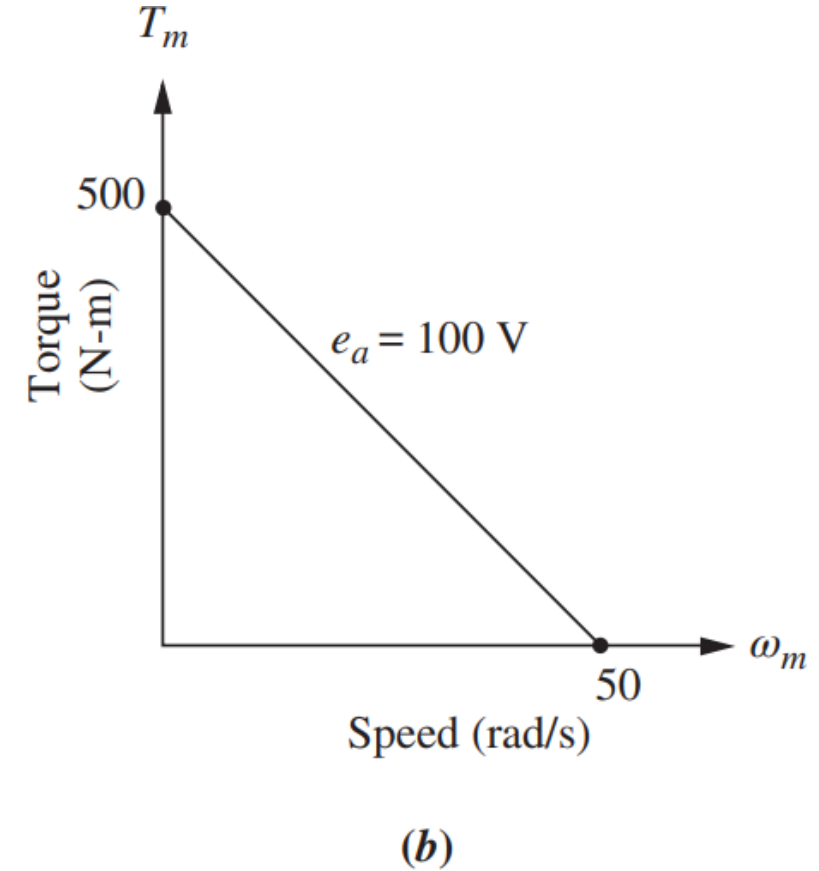
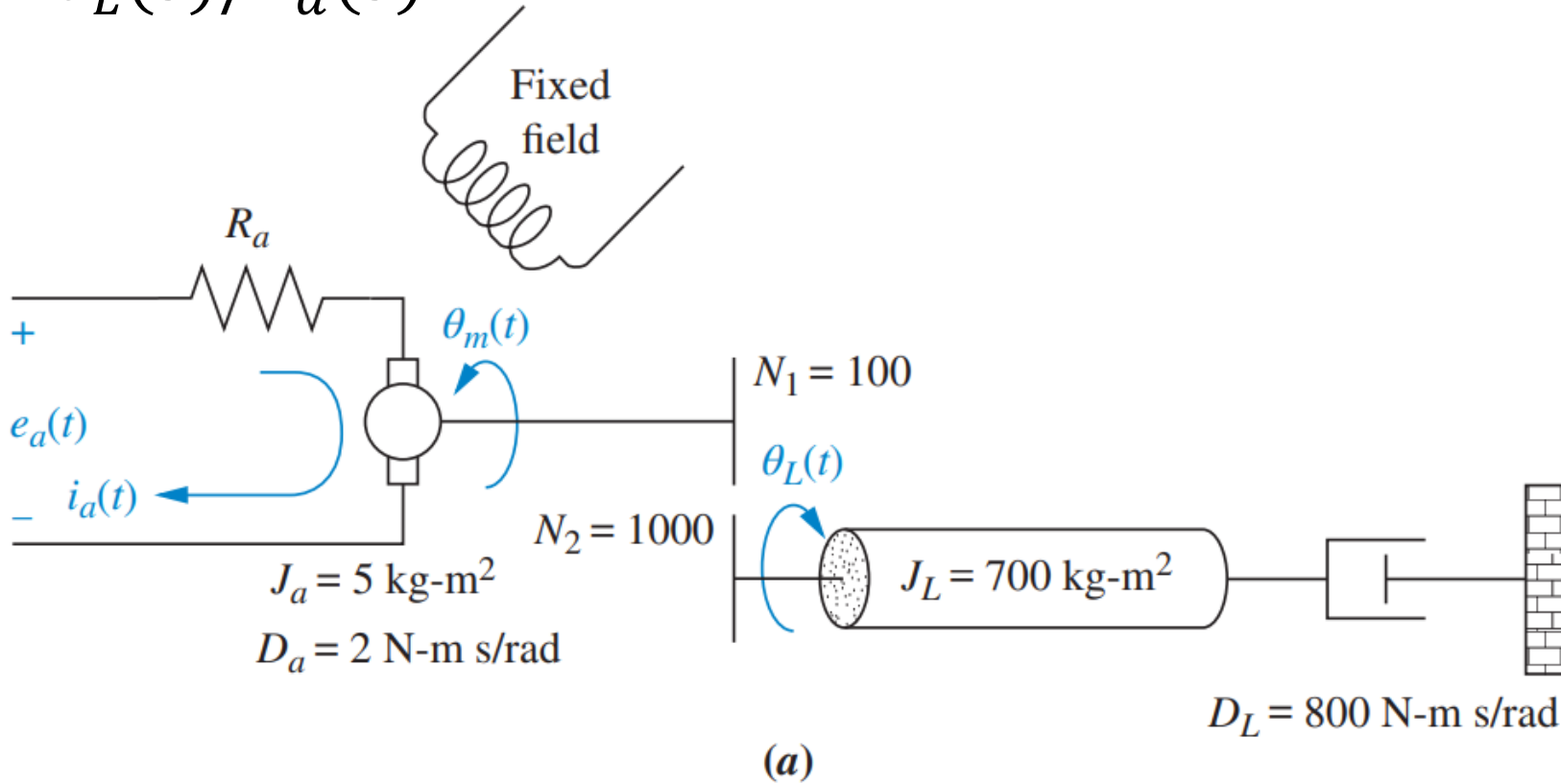


$$\frac{K_t}{R_a} = \frac{T_{stall}}{e_a}$$

$$K_b = \frac{e_a}{\omega_{no-load}}$$

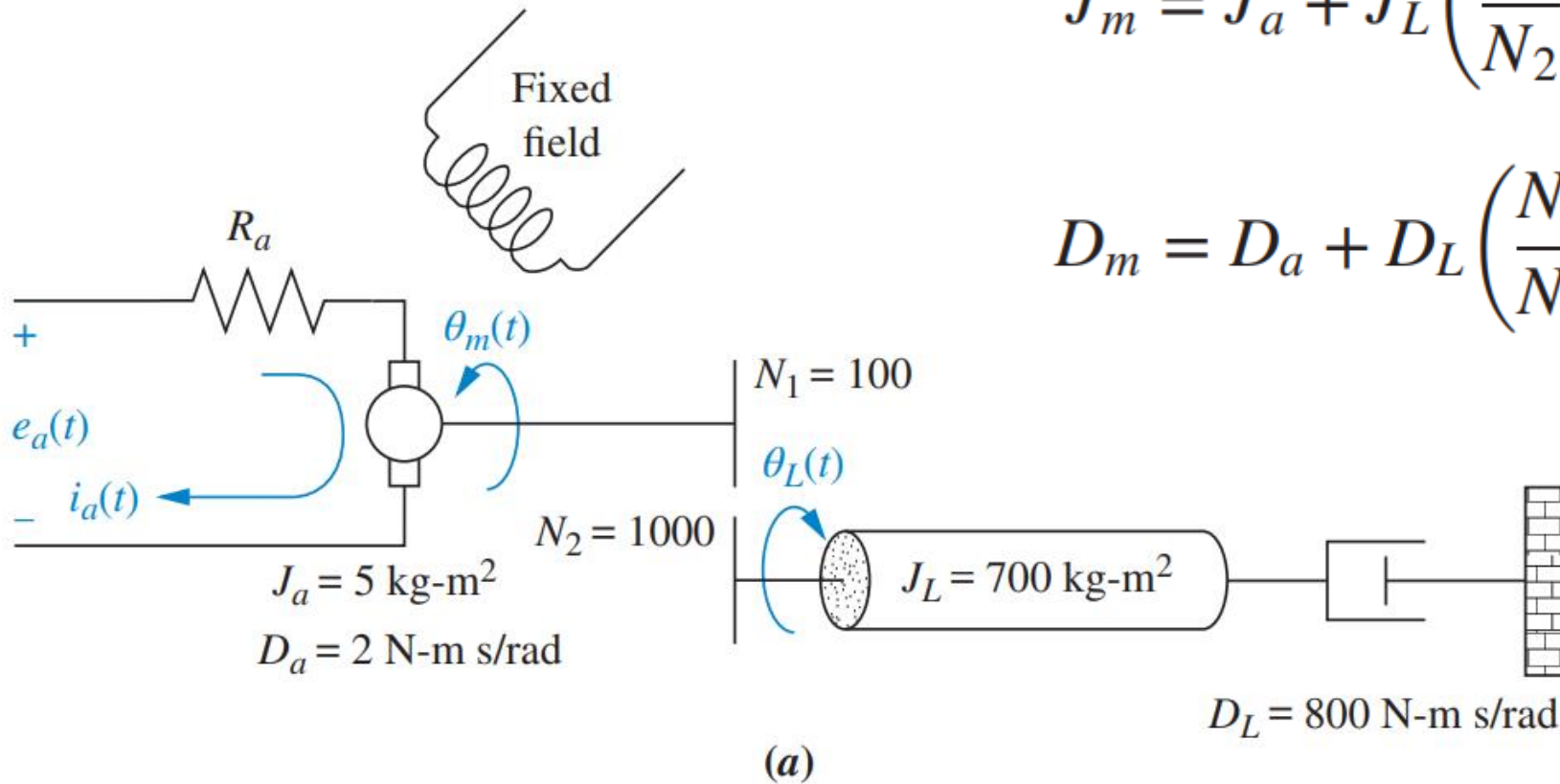
DC Servomotor (E.g.)

E.g. Given this system and torque-speed curve, find the transfer function, $\theta_L(s)/E_a(s)$.



DC Servomotor (Sln.)

E.g. Given the following system and torque-speed curve, find the transfer function, $\theta_L(s)/E_a(s)$.

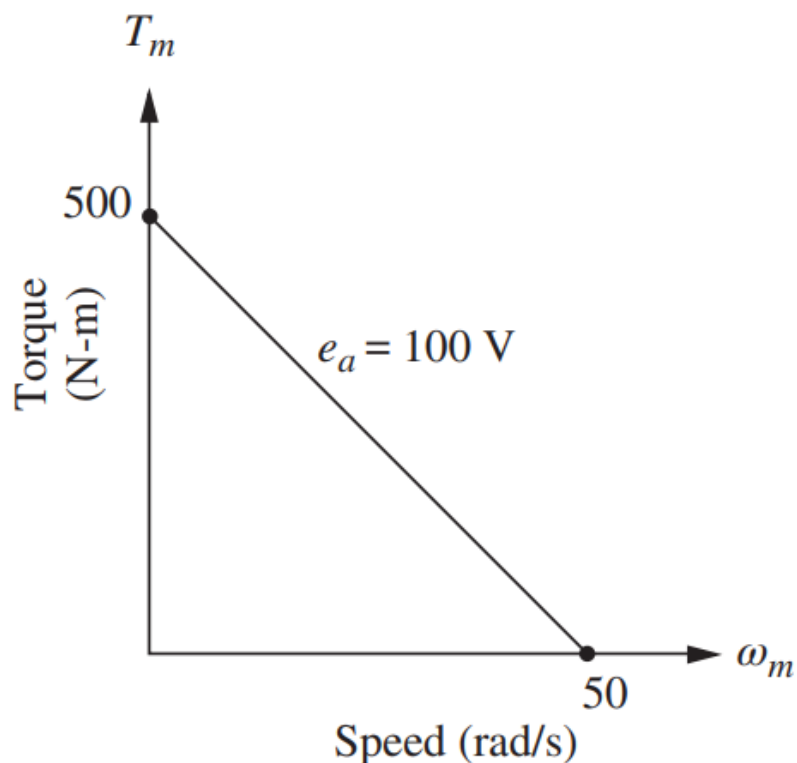


$$J_m = J_a + J_L \left(\frac{N_1}{N_2} \right)^2 = 5 + 700 \left(\frac{1}{10} \right)^2 = 12$$

$$D_m = D_a + D_L \left(\frac{N_1}{N_2} \right)^2 = 2 + 800 \left(\frac{1}{10} \right)^2 = 10$$

DC Servomotor (Sln. Cntd.)

E.g. Given the following system and torque-speed curve, find the transfer function, $\theta_L(s)/E_a(s)$.



(b)

$$T_{\text{stall}} = 500$$

$$\omega_{\text{no-load}} = 50$$

$$e_a = 100$$

$$\frac{K_t}{R_a} = \frac{T_{\text{stall}}}{e_a} = \frac{500}{100} = 5$$

$$K_b = \frac{e_a}{\omega_{\text{no-load}}} = \frac{100}{50} = 2$$

DC Servomotor (Sln. Cntd.)

E.g. Given the following system and torque-speed curve, find the transfer function, $\theta_L(s)/E_a(s)$.

Substituting:

$$\frac{\theta_m(s)}{E_a(s)} = \frac{5/12}{s \left\{ s + \frac{1}{12} [10 + (5)(2)] \right\}} = \frac{0.417}{s(s + 1.667)}$$

In order to find $\theta_L(s)/E_a(s)$, we use the gear ratio, $N_1/N_2 = 1/10$, and find

$$\frac{\theta_L(s)}{E_a(s)} = \frac{0.0417}{s(s + 1.667)}$$